



Matrix notation: $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{1n} \\ a_{21} & a_{22} & a_{23} & a_{2n} \\ a_{31} & a_{32} & a_{33} & a_{3n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & a_{mn} \end{bmatrix}$

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Student	P	C	M
A	80	85	90
B	90	80	85
C	88	87	97

$M = \text{np.array}([80, 85, 90], [90, 80, 85], [88, 87, 97])$

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ multiply with scalar (x)

Matrix multiplication: $A \rightarrow m \times n$, $B \rightarrow n \times p$, $C \rightarrow m \times p$

$C_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$

Determinants: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

3x3 matrix: $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

inverse matrix: $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$Ax = b$, $A = \begin{bmatrix} 3 & 2 \\ 5 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 1300 \\ 2300 \end{bmatrix}$

you add 2 products: $3x + 2y = 1300$, $5x + 4y = 2300$

Eigenvalues + Eigenvectors: $Ax = \lambda x$

matrix stretch: $A = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$

PCA -> dimensionality reduction

$Cv = \lambda v$, eigenvalues = principal components

SVD -> Singular value decomposition

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $A^{-1} = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$

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$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $A^{-1} = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$

$A = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} 1/5 & 0 \\ 0 & 1/2 \end{bmatrix}$